

# **CAK2HAB3 - Dasar Kecerdasan Artifisial**

## Introduction to Artificial Intelligence

Lecturer Team



# **The Very Definition of Artificial Intelligence**



**What's the first thing that comes to  
your mind when people say  
A.I.**



**If it was the 90s**







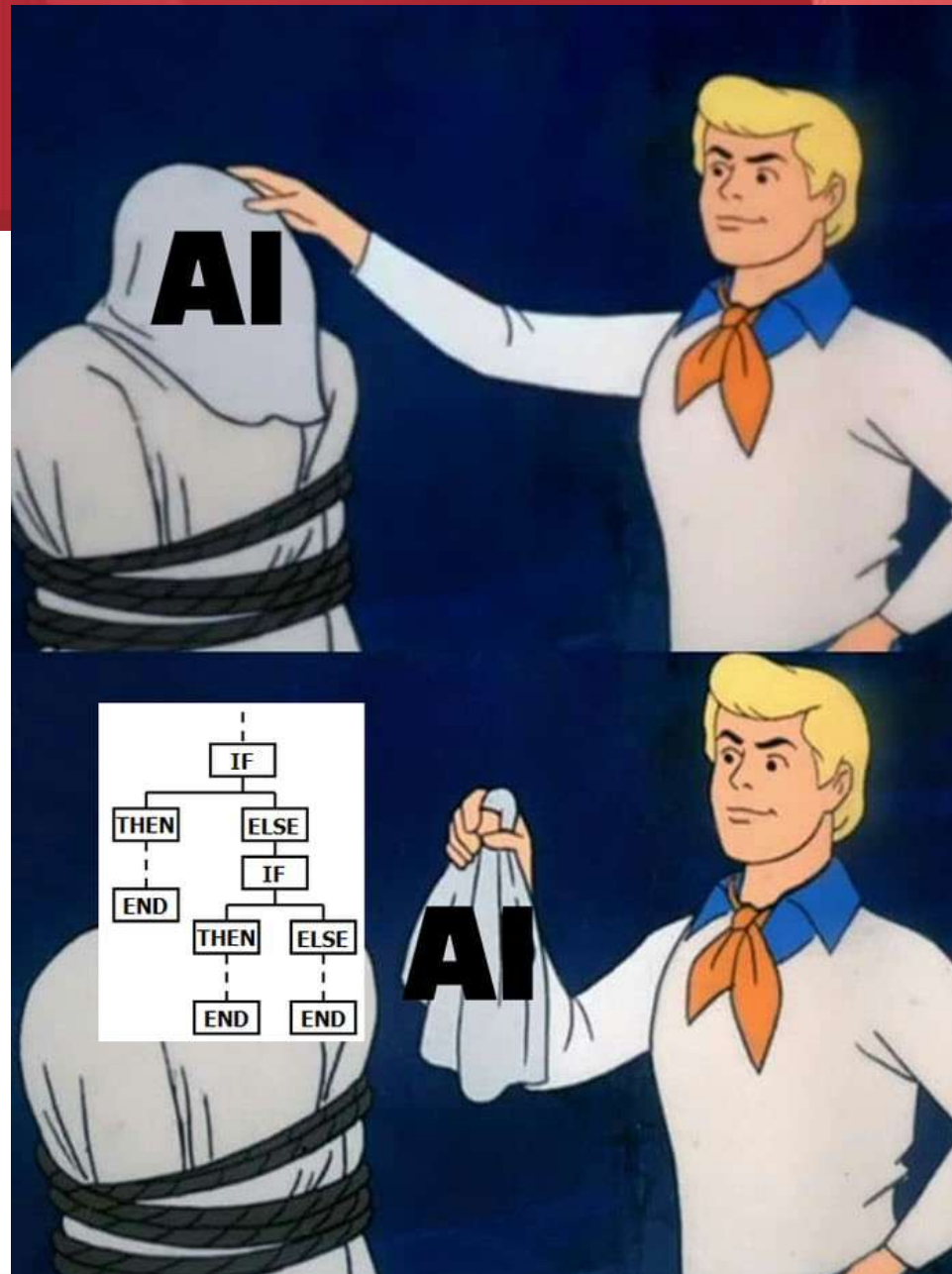
## **4 stages of AI**



## Thinking Humanly

- ▶ “The exciting new effort to make **computers think** ... machines with minds, in the full and literal sense”
  - (Haugeland, '85)
- ▶ “[The automation of] activities that we associate with **human thinking**, activities such as decision making, problem solving, learning ...”
  - (Bellman, '78)

# Thinking Humanly





## Thinking Rationally

- ▶ “The study of **mental faculties** through the use of computational models”
  - (Charniak and McDermott, '85)
- ▶ “The study of the computations that make it possible to **perceive, reason, and act**”
  - (Winston, '92)

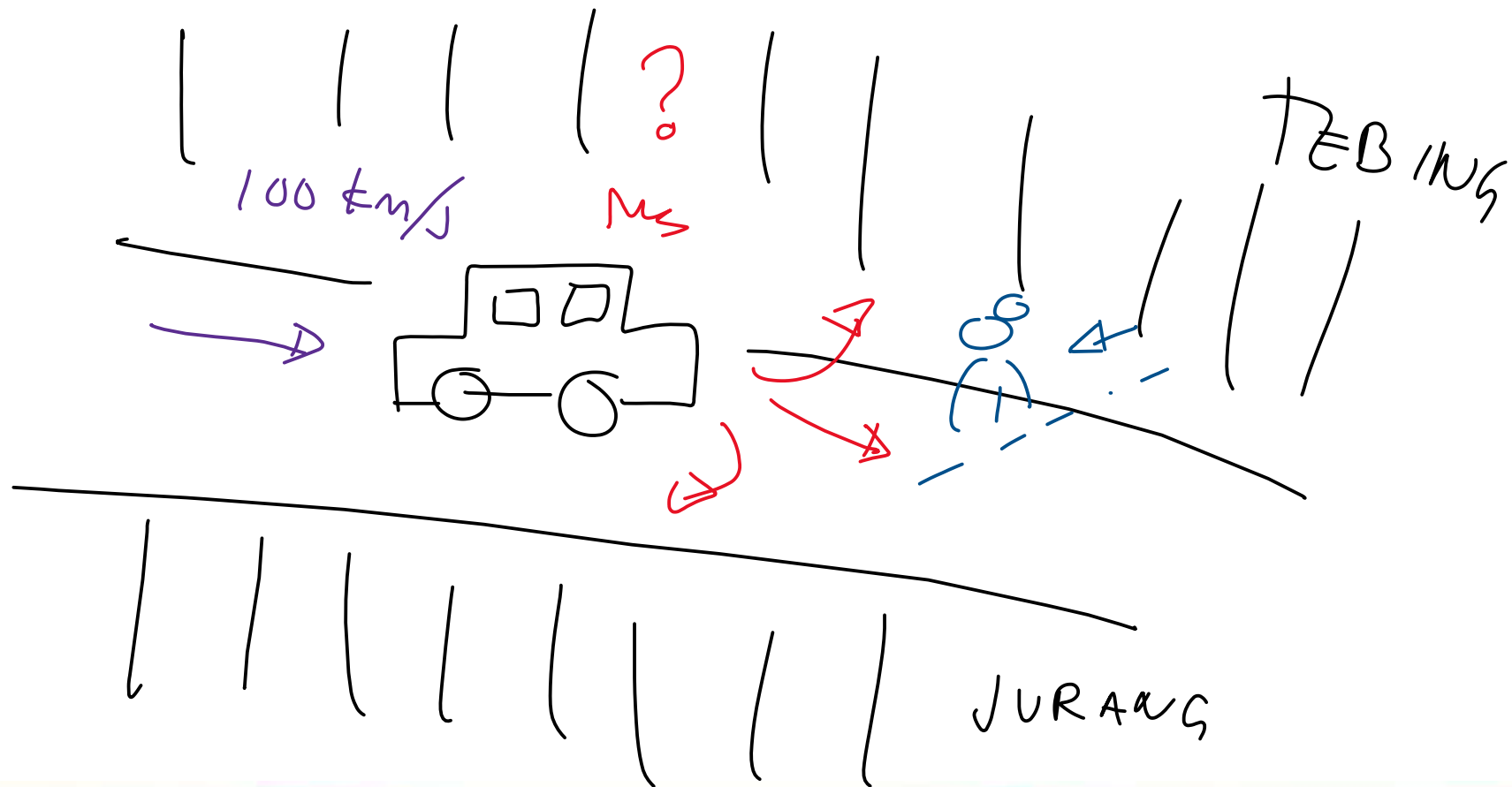
## Acting Humanly

- ▶ “The art of creating machines that perform functions that require **intelligence** when performed by people”
  - (Kurzweil, '90)
- ▶ “The study of how to make **computers do things** which, at the moment, people do better”
  - (Rich and Knight, '91).

## Acting Rationally

- ▶ A field of study that seeks to explain and emulate **intelligent behavior** in term of computational processes”
  - (Schalkoff, '90)
- ▶ “The branch of computer science that is concerned with the automation of **intelligent behavior**”
  - (Luger and Stubblefield, '93)

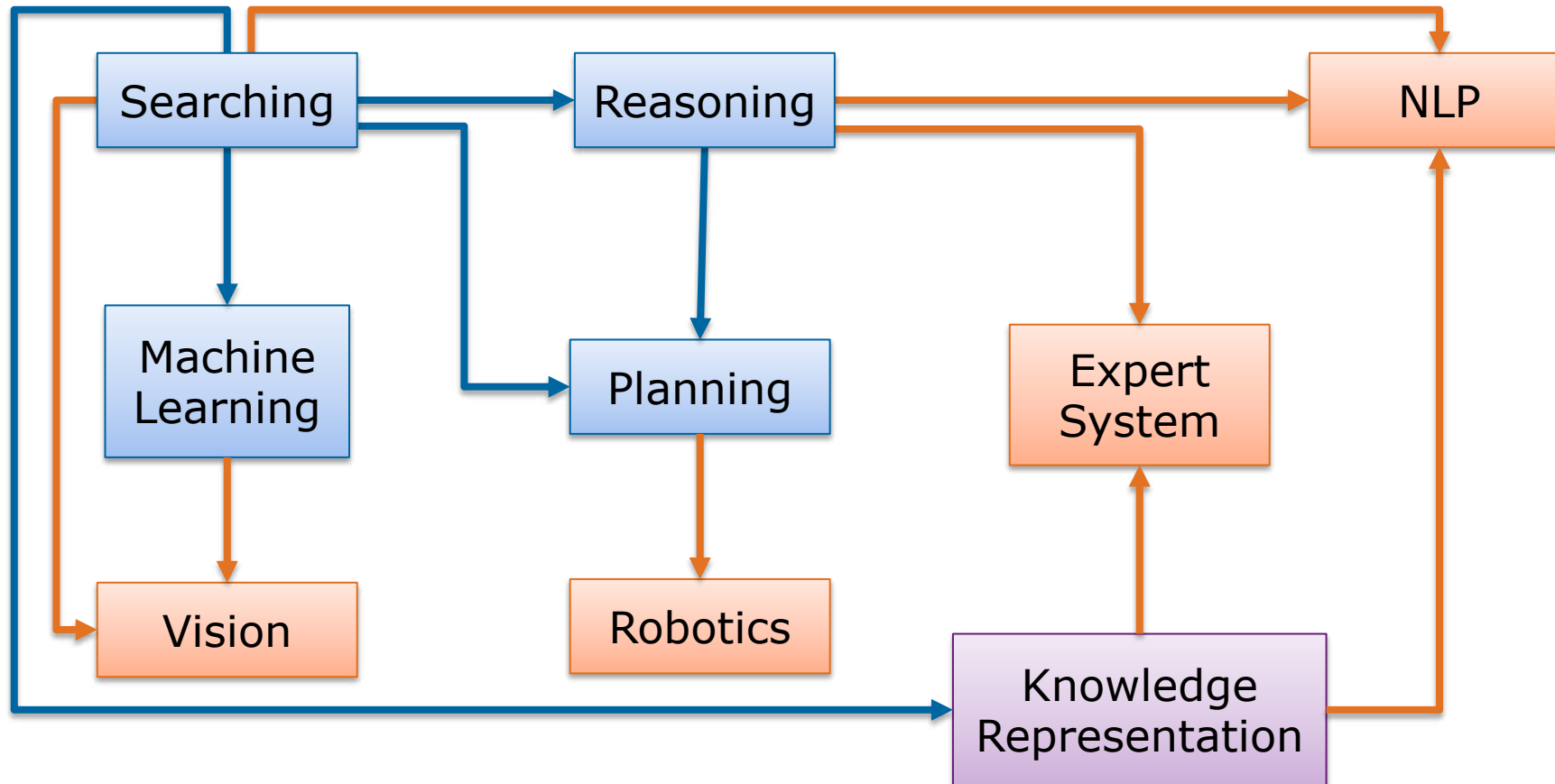
## Acting Rationally



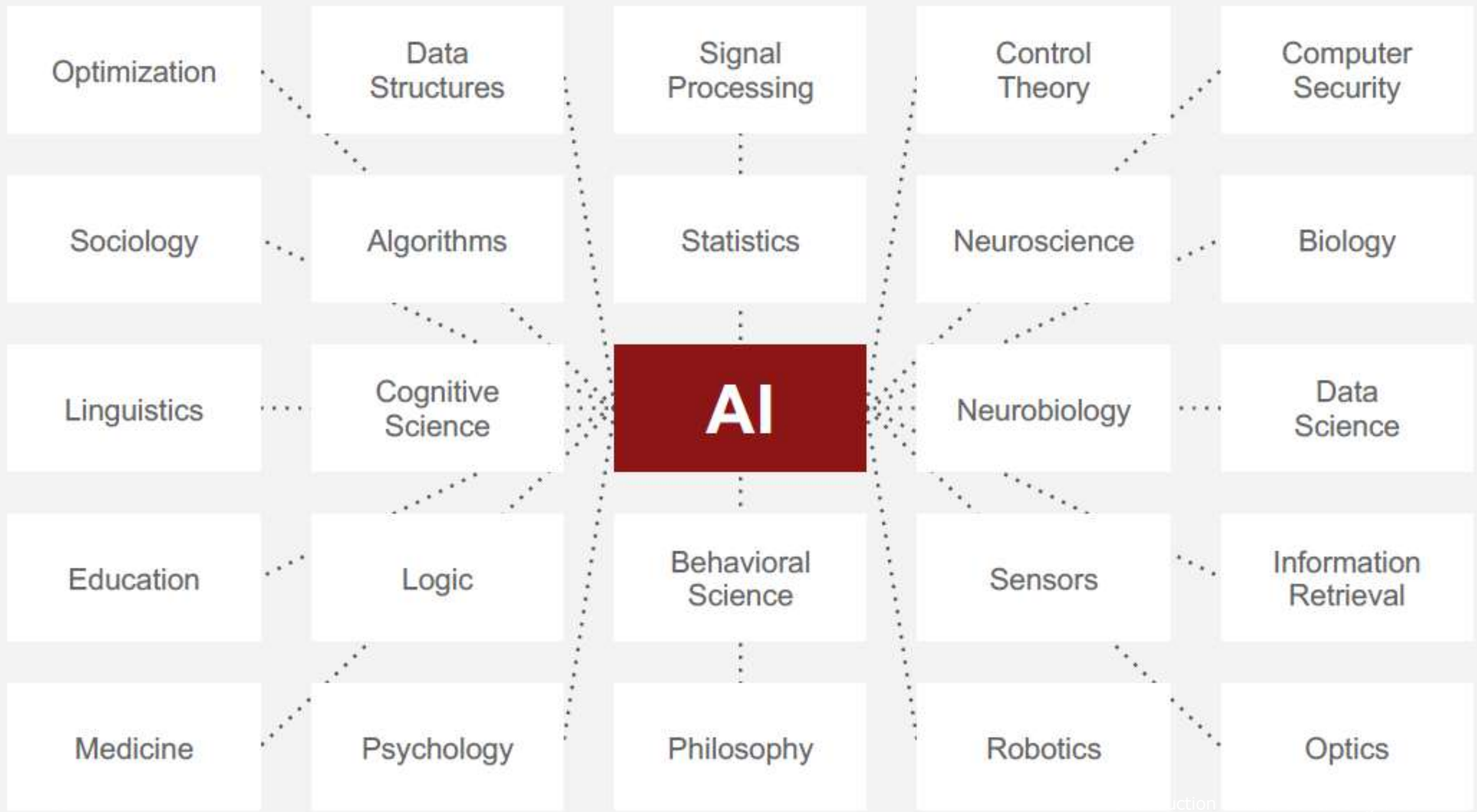


# **Application of Introduction to Artificial Intelligence**

## Some Areas of AI applications





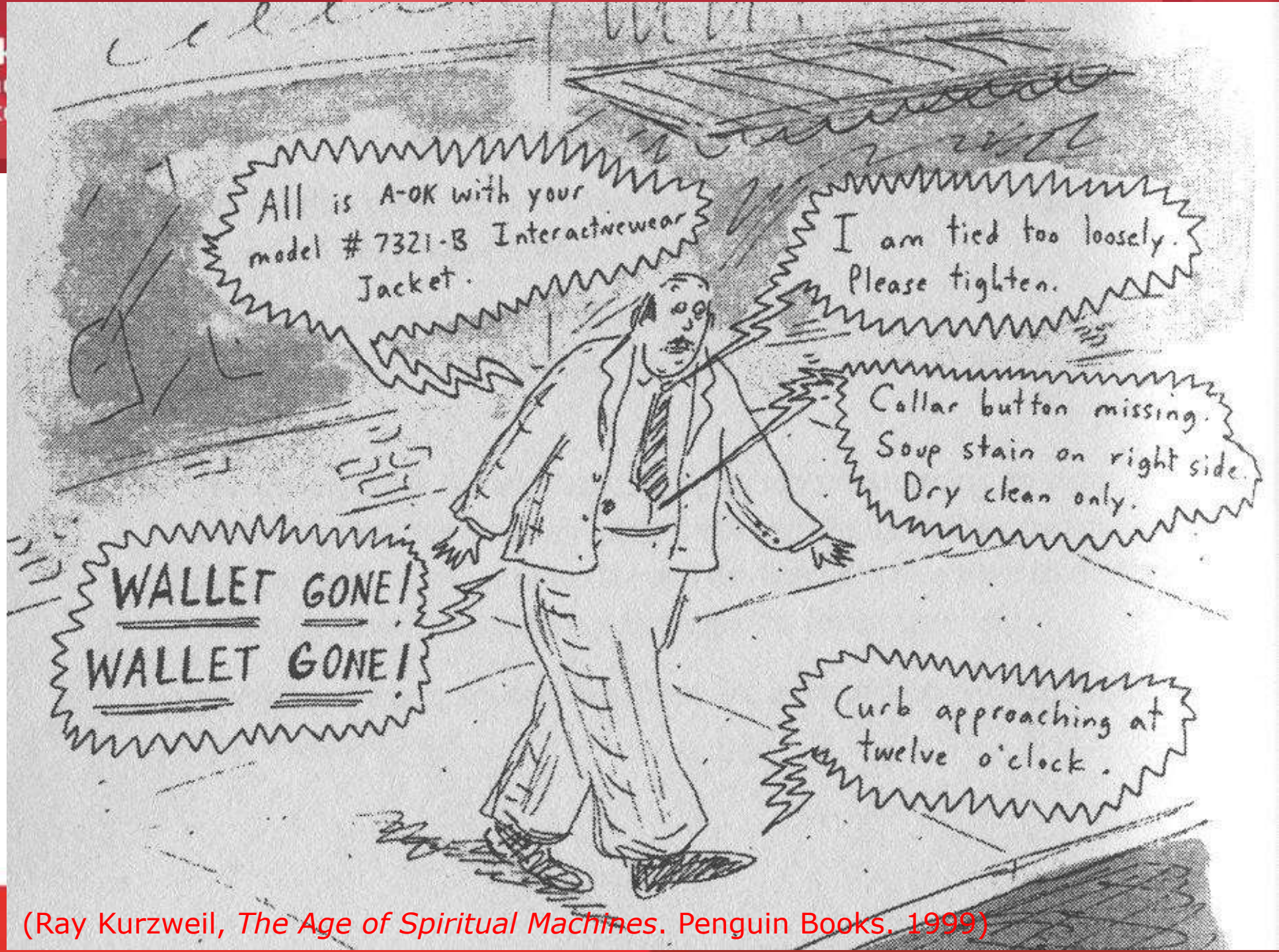




## **AI Application**

- Navigation System
- Visual Recognition/Understanding
- Medical Records
- Automatic Control System
- Data Mining and Filtering
- Speech Recognition
- Game Development









## Break, Question?



# **History of AI**

## **- the ups and downs -**

## History of AI

### ► UNIVAC I (1951)

- the first general purpose electronic digital computer designed for business application
- Accurately predicting the 1952 US presidential election



50s

## History of AI

- ▶ AI Early Developments (1952 – 1969)
  - Newell & Simon → General Problem Solver
  - 1958: McCarthy → LISP
  - McCarthy → Programs with Common Sense
  - 1957: [Frank Rosenblatt](#) → Mark I Perceptron



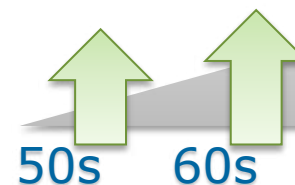


# History of AI

## ▶ AI Early Developments (1952 – 1969)

– Eliza Chat bot (1966)

- E : what is bothering you?
- A : I feel sad all the time
- E : ( recognize "I feel-" ) Why do you feel sad all the time?
- A : I don't get along with my friends
- E : ( recognize "my friends" ) Tell me about your friends
- A : ...



# History of AI

## ▶ AI Early Developments (1952 – 1969)

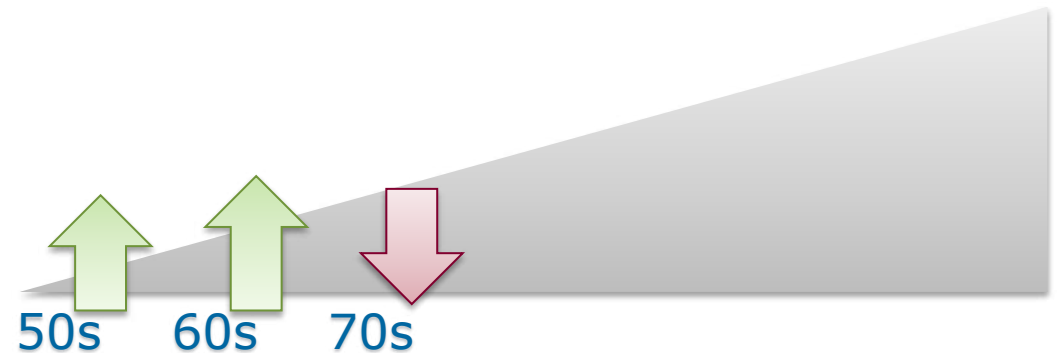
- AI Hype through Blockbuster movies
- 2001: A Space Odyssey (1968)



## History of AI

### ► Slow (poor) Developments (1966 - 1974)

- The AI program contains little knowledge of the subject and succeeds simply because of simple synthetic manipulations.
- Hard problem to solve → AI programs fail.

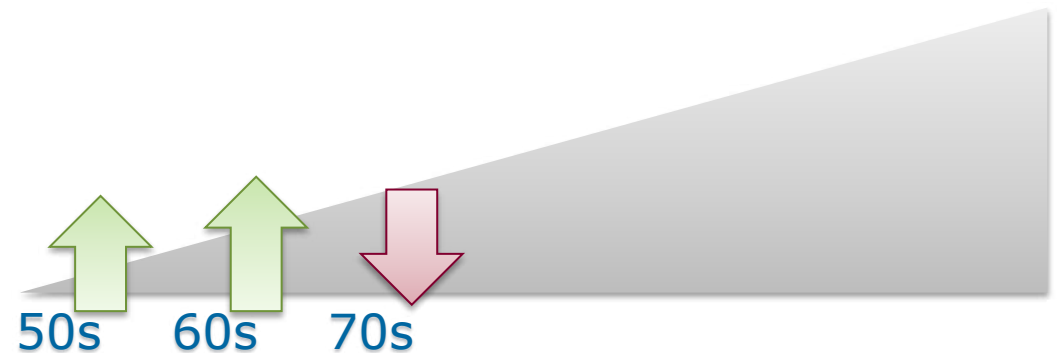


## History of AI

### ► Slow (poor) Developments (1966 - 1974)

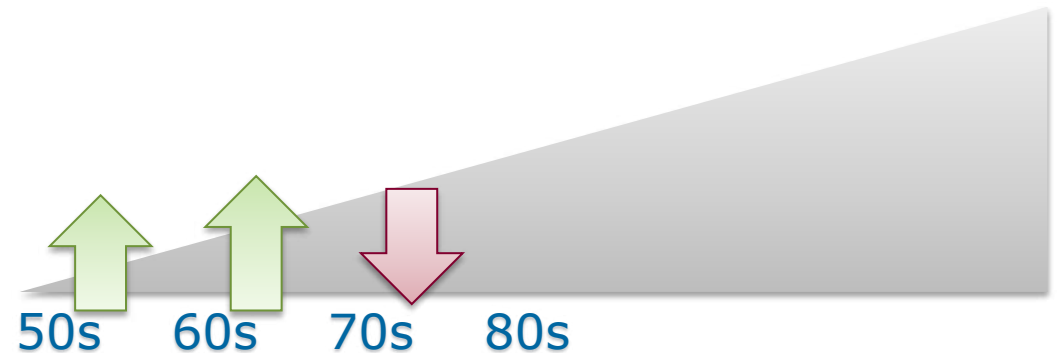
- There are some limitations on the basic structure used to generate intelligence behavior.
- Two different perceptron entries can not be trained to recognize the two different entries.

### ► First AI Winter



## History of AI

- ▶ Knowledge-based System ( 1969 – 1979 )
- ▶ AI entering industrial (1980 - 1988)
  - Expert system
  - Digital Equipment
  - Factory robots

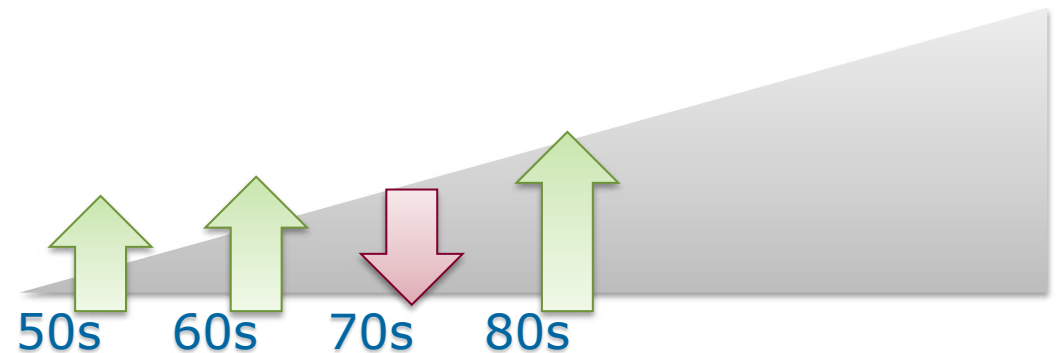
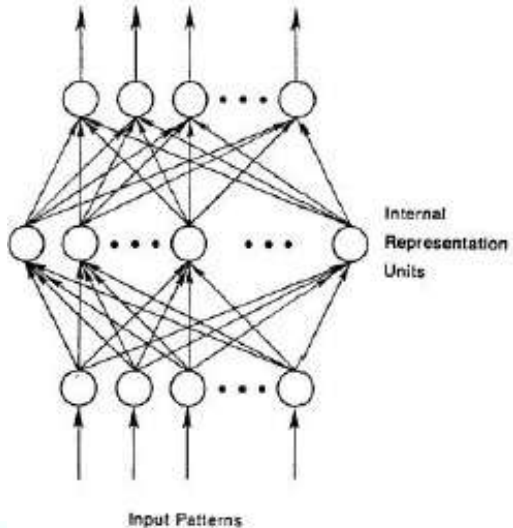




# History of AI

## ► The Return of Artificial Neural Networks (1986~)

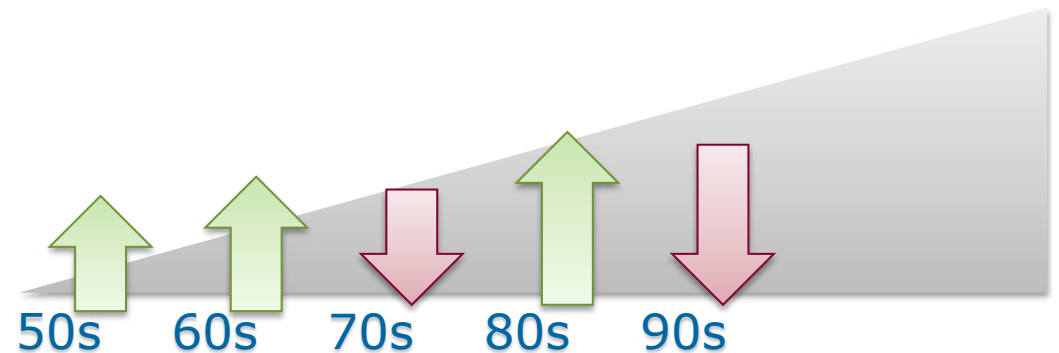
- 1985s: Back-Propagation learning algorithm introduced
- Used in computer science and psychology



# History of AI

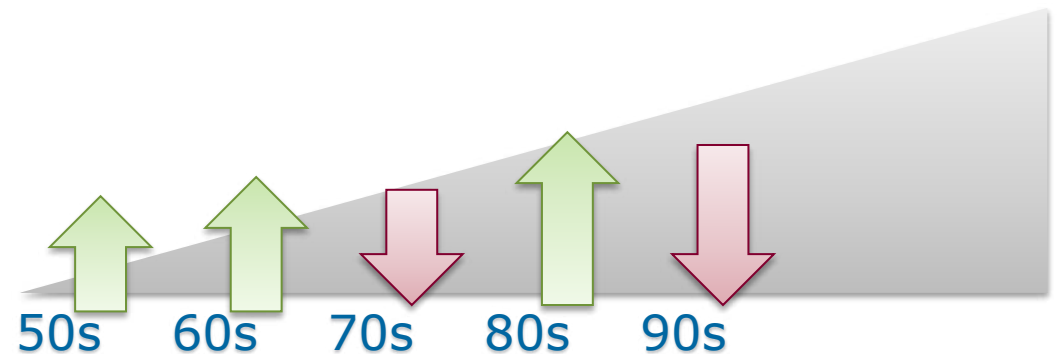
## ► Neural Networks Limitation (1990~)

- Limited data availability
- Limited computational power
- Failures identified in Neural Networks  
(Vanishing gradient problems)



# History of AI

- Second AI Winter
- Era of Feature Extractions

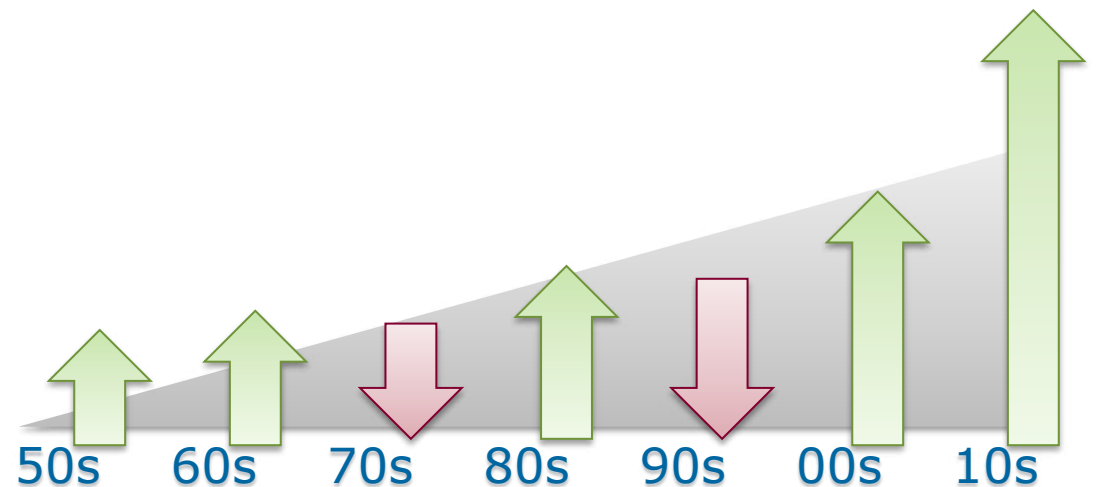


## History of AI

### ➤ Geoff Hinton's Deep Learning (2006)

- mathematically optimize results from each layer so that the learning accumulated faster as it proceeded up the stack of layers
- AutoEncoder, Restricted Boltzmann Machine

### ➤ AI boom (2011~)





# **Today's AI**

## **Today's AI**

- ▶ [2011] IBM's Watson won a game of Jeopardy against the top two all-time champions
- ▶ [2015-2018] Neural Net beat human accuracy in 1000-class Visual Recognition in IMAGENET
- ▶ [2016, 2017] Google DeepMind's AlphaGo beat 2 go masters





## **Todays AI**

- ▶ [2017] DeepBlue beat poker pro players
- ▶ [2017] Elon Musk's OpenAI Dota2 bot beat pro player in 1-vs-1
- ▶ [2019] Elon Musk's OpenAI Dota2 bot beat pro player in 5-vs-5
- ▶ [2018] A portrait created by AI just sold for \$432,000.



## **Today's AI**

- ▶ AI can paint
  - Deep Dream, Generative Adversarial Network
- ▶ AI can write a script and screenplay for a movie
  - Sunspring
- ▶ AI can compose a music
  - AIVA
- ▶ AI can generate (write) a code
  - DeepCoder



## **Today's AI**

- ▶ Self driving car
- ▶ Smart Home
- ▶ Automatic detection and recognition
  - Face, fingerprint, sound, speech
- ▶ Real speech-to-speech time translation
- ▶ Image captioning and understanding
- ▶ Text summarization

## == ARTIFICIAL INTELLIGENCE AT A GLANCE 2018 ==



Just **15%**  
of organizations use AI  
today; by next year, that  
number will be 31%

[cmo.com/features/articles/2018/2/26/adobe-2018-digital-trends-report-findings.html#gs.kqGOKgc](https://cmo.com/features/articles/2018/2/26/adobe-2018-digital-trends-report-findings.html#gs.kqGOKgc)



AI-focused startups  
have seen a  
**14-fold**  
increase since 2000

[cdn.aindex.org/2017-report.pdf](https://cdn.aindex.org/2017-report.pdf)

Consumers often don't know  
when they are interacting  
with AI systems, as only  
one-third of them said they  
use an AI-powered service or  
device, when the number  
who actually do is

**77%**

[pega.com/system/files/resources/2017-11/what-consumers-really-think-of-ai-infographic.pdf](https://pega.com/system/files/resources/2017-11/what-consumers-really-think-of-ai-infographic.pdf)



Global spending on AI is  
projected to surge to more than  
**\$7 billion**  
per year by 2022, more than double  
the \$2 billion spent this year

[businesswire.com/news/home/20180131005068/en/juniper-research-retailer-spending-ai-grow-fourfold](https://businesswire.com/news/home/20180131005068/en/juniper-research-retailer-spending-ai-grow-fourfold)





## **Future of AI ?**

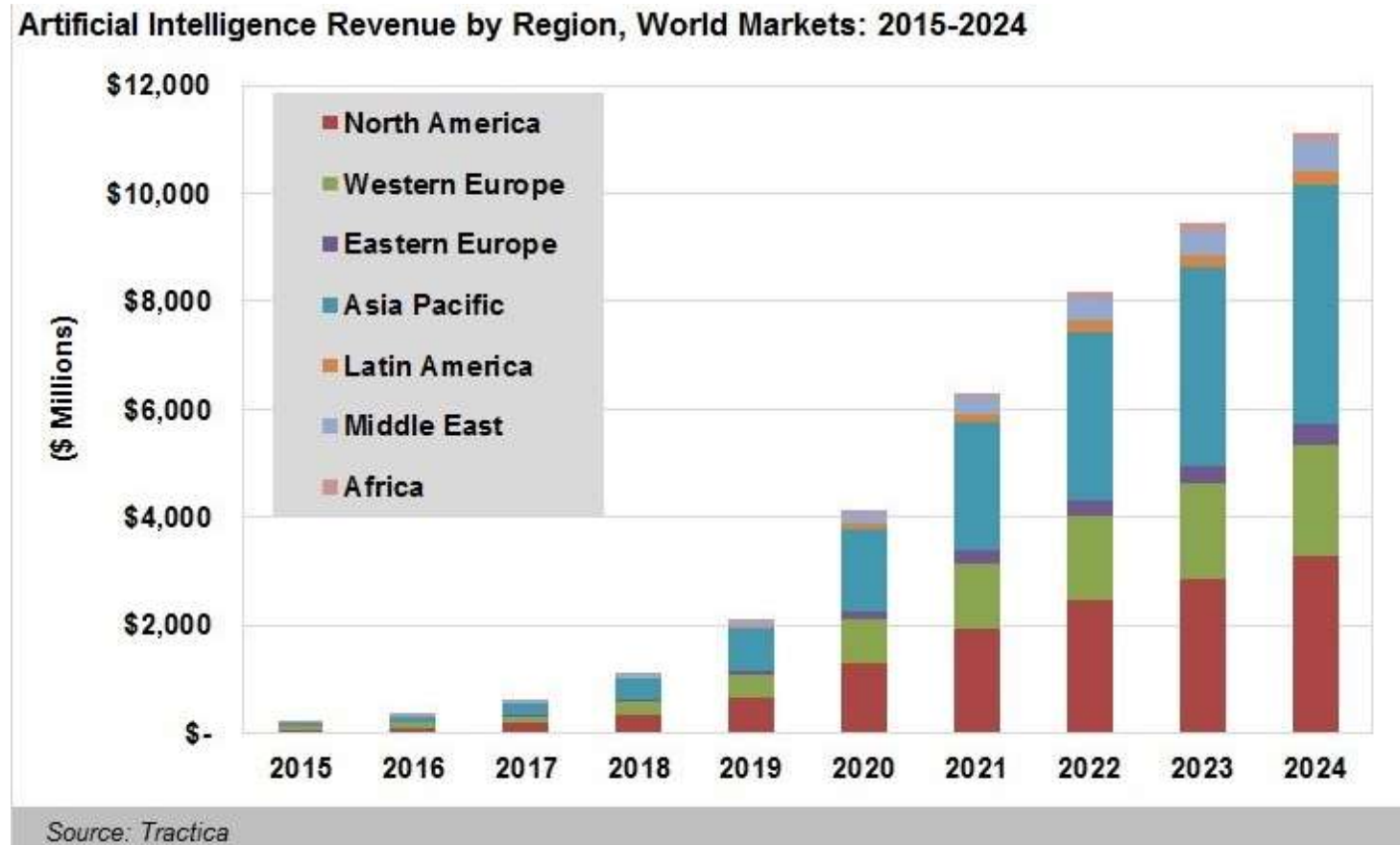


## **Future of AI**

- AI is the new electricity
- Computing is becoming free
- Data is becoming the new oil
- Machine learning is becoming the new combustion engine



# Future of AI



# Future of AI





# Question?



## Next Agenda

- ▶ Introduction to Searching



## References

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*THANK YOU*